

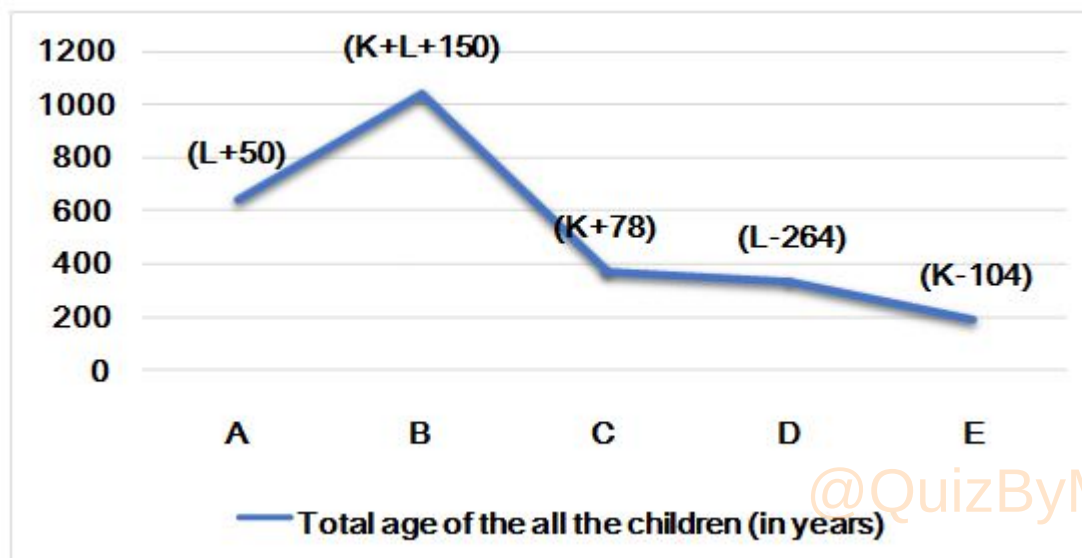
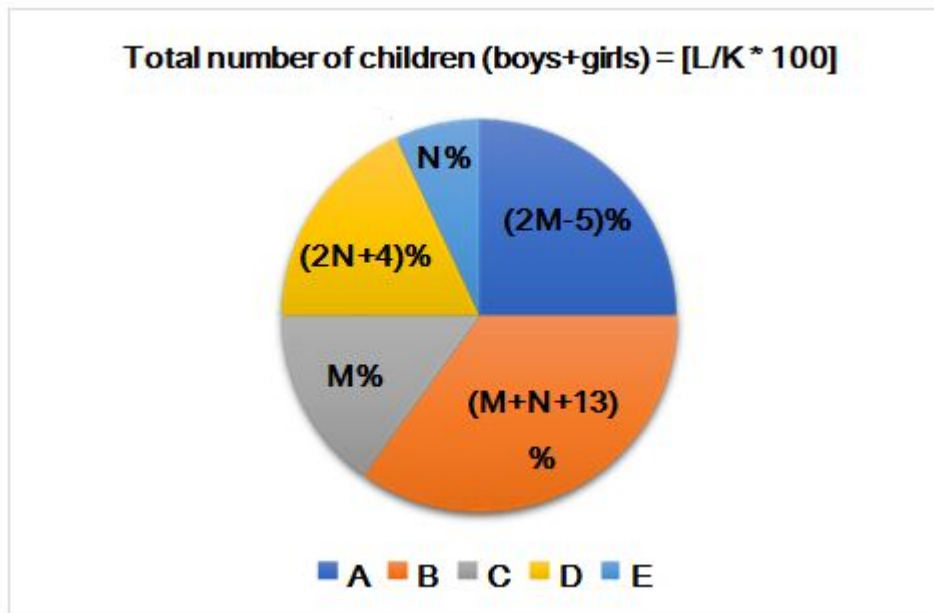
1. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of children (boys + girls) living in five different towns A, B, C, D, and E.

The line graph given below shows the total age of all the children (boys + girls) living in each town.

The total number of children (boys + girls) living in all the five towns together is $[L/K * 100]$.



Note:

The values of M and N are positive integers.

The value of N is a prime number, which lies between both the roots of the equation $Z^2 - 13Z + 36 = 0$.

The value of M is an odd composite number.

The ratio of the values of M to that of K is 1:20.

The value of L is 100 less than the value of (10^2N) .

The total number of children living in any town = the number of boys living + the number of girls living

The total age of all the children living in any town = {(the total number of children) $\times$ (the average age of each child)}

In town A, if the average age of all the boys is 15 years, which is 50% more than that of girls, then find the difference between the number of boys and girls living in town A.

- a. 10 years
- b. 16 years
- c. 13 years
- d. 19 years
- e. None of these

2. Questions

In town B, the ratio of the number of boys to that of girls is 4:3 and the average age of each girl is 14 years less than that of each boy. Find the sum of the total age of all the girls living in town B.

- a. 270 years
- b. 250 years
- c. 200 years
- d. 210 years
- e. None of these

3. Questions

In town C, if there are adults (males + females) are also living, which is 70% of the total number of people (children + adults) living in town C, then find the number of males living in town C, if it is $\frac{5}{7}$ th of the total number of adults (males + females) living in the same town.

- a. 20
- b. 50
- c. 70
- d. 30
- e. None of these

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4. Questions

If there is another town F, where the total number of children (boys + girls) living is 11.11% more than that of town D and the average age of all the boys is 8 years, which is twice the average age of

all the girls in the same town. Find the ratio of the number of boys to that of girls living in town F, if the total age of all the boys is 140 years more than that of girls in the same town.

- a. 2:1
- b. 4:5
- c. 3:5
- d. 3:2
- e. None of these

5. Questions

In town E, if the product of the number of boys and girls is 49, then find the average age of all the boys, if the average age of all the girls is 14 years.

- a. 14 years
- b. 21 years
- c. 7 years
- d. Cannot be determined
- e. None of these

6. Questions

Read the following information carefully and answer the questions based on it.

The table given below shows the total number of players playing three different sports i.e., cricket, football and hockey from five different sports clubs P, Q, R, S and T. It also shows the percentage of the number of players playing football out of the total number of players playing all three sports together in each sports club. The table also shows information about the difference between the number of players playing cricket and hockey (cricket-hockey) from each sports club in the form of a statement.

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Club	The total number of players playing (cricket + football + hockey)	The percentage of the number of players playing football	The difference between the number of players playing cricket and hockey (cricket-hockey)
P	$(A+20)$	40%	10% of the total number of players playing all the sports
Q	210	2B%	42 less than the number of players playing football
R	$(C+90)$	C%	$\frac{2}{3}$ rd of the number of players playing cricket
S	20D	55%	Equal to the number of players playing hockey
T	240	$(6E+3)\%$	The square root of the difference between the number of players playing cricket and football (football-cricket)

Note:

The values of the variables are positive integers.

In each sports club, the number of players playing football is more than that of hockey.

In club P, the number of players playing hockey is 80.

The ratio of the number of players playing football to that of hockey in club Q is 5:1.

The number of players playing football in club R is 90.

The number of players playing cricket in club S is 50% more than the number of players playing hockey in club P.

In club T, the number of players playing football is 180, whereas the number of cricket players playing is 3E.

The number of players playing football is more than that of cricket in club T.

The total number of players playing = the number of players playing cricket + the number of players playing football + the number of players playing hockey

In club P, out of the total number of players playing cricket, the ratio of the number of average runs scored by 44 players and 38 players is 5:4. If the average runs scored by the remaining number of players is 30 and the total number of runs scored by all the players playing cricket is 2760, then find the difference between the total number of runs scored by 44 players and 38 players.

- a. 420
- b. 260
- c. 180
- d. 340
- e. None of these

7. Questions

If there is another sports club U, where the total number of players playing all the sports is 40 less than that of sports club T and the number of players playing cricket is X% of the total number of players playing all the sports. If the ratio of the number of players playing football to that of hockey is 13:7 and the number of players playing cricket is 35 more than that of football, then find the total number of players playing hockey in sports club U.

- a. 41
- b. 35
- c. 23
- d. Cannot be determined
- e. None of these

8. Questions

The ratio of the number of players playing football above 25 years of age from club Q to that of players playing football below 25 years of age from club R is 12:7. While the number of players playing football below 25 years of age from club Q is $\frac{2}{11}$ less than the number of players playing football above 25 years of age from club R. Find the total number of players playing football above 25 years of age is what percent of the total number of players playing football below 25 years of age from both the clubs together?

- a. 143.75%
- b. 122.22%
- c. 112.50%
- d. 140.67%
- e. None of these

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9. Questions

If there is another sports club S i.e., volleyball and the number of players playing volleyball from club S is [K] more than the number of players playing hockey from the same club and is also [L] less than the number of players playing cricket from the same club, then find the value of $\sqrt{(K+L+4)}$.

- a. 4
- b. 16
- c. 8
- d. 32
- e. None of these

10. Questions

In club T, if the average age of all the players playing hockey is $[J]$ years and if the age of their coach is also included then the average age of all the players including their coach increases by 0.80 years. If the age of their coach is $[2J]$ years, then find the value of $[J^2]$

- a. 424
- b. 400
- c. 441
- d. Cannot be determined
- e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

A certain number of employees (males + females) work in two different banks A and B. Out of the total number of employees in each bank, some are officers whereas the rest are clerks.

Bank A: The ratio of the number of male officers working to that of female clerks working in the bank is 3:1. Whereas, the number of male clerks working in bank A is $\frac{1}{7}$ th less than the number of female officers working in the same bank. The total number of male (officers + clerks) employees working in bank A is 150, which is 50% more than that of female (officers + clerks) employees.

Bank B: In bank B, the total number of male (officers + clerks) employees working is 140, which is 80 more than that of female (officers + clerks) employees working in the same bank. The number of male officers working in bank B is 150% more than that of female officers working in the same bank. The ratio of the number of male to female clerks working in bank B is 2:1.

Note:

The total number of employees working = the number of male employees working + the number of female employees working

The total number of employees working = the number of officers working + the number of clerks working

If out of the total number of male (officers + clerks) employees working in bank A, the ratio of the number of senior officers to that of junior clerks working is 8:7. The number of junior officers working in bank A is twice the number of senior clerks working in the same bank.

Find the correct relation between quantity I and quantity II.

Quantity I: The difference between the number of senior officers and clerks working in bank A.

Quantity II: The average number of junior officers and clerks working in bank A.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

12. Questions

In bank B, out of the total number of female clerks working, the difference between the number of graduates and post graduates working is $[K^2]$, while the number of post graduates working is $[4K]$ and the number of graduates working is more than that of post graduates working in bank B among female clerks. Find the value of $[\sqrt{L}]$, if $2K = L$. (K is a positive integer)

- a. 2
- b. 5
- c. 3
- d. Cannot be determined
- e. None of these

13. Questions

If there is another bank C, where the total number of employees working is the average of the total number of employees working in bank A and bank B together. If the total number of clerks working in bank C is 33.33% more than that of bank A, then find the total number of officers working in bank C.

- a. 100
- b. 80
- c. 105
- d. 75
- e. None of these

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14. Questions

There are three quadratic equations I, II and III given below

Equations	Remarks
I). $X^2 - 41X + P^2 = 0$	c and d are the roots of the equation, where both c and d are exact multiples of 5 and also the difference between the roots of c and d is 5. ($c > d$)
II). $Y^2 - 35Y + 2Q = 0$	e and f are the roots of the equation, where both the roots are prime numbers. ($e = f$)
III). $Z^2 - 34Z + (12R+1) = 0$	a and b are the roots of the equation, where both a and b are the squares of a positive integer. ($a > b$)

Find the value of $[(Q-50)/(R-P)]$.

- a. 49
- b. 25
- c. 36
- d. 64
- e. None of these

15. Questions

There are three quadratic equations I, II and III given below with their roots as X, Y and Z but not necessarily in the same order.

Equation	Roots
I). $A^2 + A - M = 0$	X). 30 and 23
II). $B^2 - 2B - 3N = 0$	Y). 20 and -21
III). $C^2 - 53C + 2K = 0$	Z). -7 and 9

Find the correct pair of equations with its roots.

- a. I-Y
- b. II-X
- c. III-Z
- d. All of the above
- e. None of these

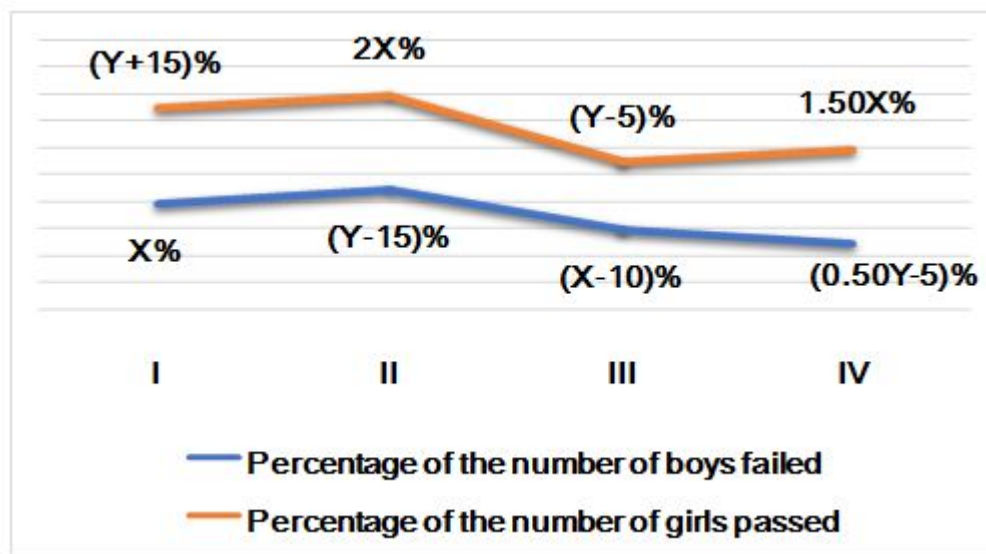
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16. Questions

Read the following information carefully and answer the questions based on it.

The line graph given below shows the percentage of the number of boys who failed out of the total number of boys who appeared in an examination from four different classes i.e., class I, class II, class III and class

IV. It also shows the percentage of number of girls who passed out of the total number of girls who appeared in the same examination from each class.



Note:

In class I, the ratio of the number of boys who passed to those of failed the examination is 3: 2, whereas the number of girls who failed the examination is 66.67% less than that of the number of girls who passed the examination.

The total number of students = the number of boys + the number of girls

The total number of students in the class = the number of students who appeared + the number of students who did not appear.

The total number of students who appeared in the examination = the number of students who passed + the number of students who failed

Out of the total number of students (boys + girls) who appeared in the examination from class III, 65% passed the examination. Find the number of girls who failed the examination out of the total number of girls who appeared in the examination from class III, if the total number of students (boys + girls) who appeared in the examination is 360.

- a. 45
- b. 54
- c. 66
- d. Cannot be determined
- e. None of these

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17. Questions

In class II, if the number of boys who passed the examination is 88 and the total number of girls (passed + failed) who appeared in the examination is 15/23 times the total number of students (boys + girls) who appeared in the same examination from the same class, then find the total number of students (boys + girls) who appeared in the examination from class II.

- a. 460
- b. 340
- c. 290
- d. 150
- e. None of these

18. Questions

If the difference between the number of boys who passed and failed the examination from class IV is 90, then find the total number of girls who passed the examination from the same class, if the number of girls who failed the examination is 55 more than that of boys who failed the examination from class IV.

- a. 100
- b. 180
- c. 150
- d. 120
- e. None of these

19. Questions

If there is another class V, if the total number of students (boys + girls) who appeared is 400 and the ratio of the number of boys to that of girls who appeared in 5:3. If the number of boys who passed the examination is 6 times the number of girls who failed the examination, whereas the number of boys who failed the examination is 41.67% less than that number of girls who passed the examination. Find the total number of students (boys + girls) who passed the examination from class V.

- a. 300
- b. 250
- c. 200
- d. 350
- e. None of these

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20. Questions

Out of the total number of students in the class, 15% did not appear in the examination. If the number of boys who passed the examination from class I is 120, which is 85 more than the number of girls who failed the examination from the same class, then find the total number of students (appeared + did not appear in class I).

- a. 144

- b. 256
- c. 324
- d. 400
- e. None of these

Explanations:

1. Questions

$$2M - 5 + M + N + 13 + M + 2N + 4 + N = 100$$

$$M + N = 22 \text{ ----- (i)}$$

From the given equation, we have,

$$Z^2 - 13Z + 36 = 0$$

$$Z^2 - 9Z - 4Z + 36 = 0$$

$$Z(Z-9) - 4(Z-9) = 0$$

$$(Z-4)(Z-9) = 0$$

$$Z = 4 \text{ or } 9$$

So, the roots are = 4 and 9

Therefore, the value of N will be between 4 and 9.

So, the possible values of N = 5 or 7 (only prime numbers between 4 and 9)

Let, the value of N = 5

Now, put the value of N in equation (i), we get,

$$M + N = 22$$

$$M + 5 = 22$$

$$M = 17 \text{ (not possible)}$$

17 is a prime number, but not an odd composite number.

Now, let the value of N = 7

Put the value of N in equation (i), we get,

$$M + N = 22$$

$$M + 7 = 22$$

$$M = 15 \text{ (possible)}$$

15 is an odd composite number.

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Therefore, the values of,

$$N = 7 \text{ (confirmed)}$$

$$M = 15 \text{ (confirmed)}$$

$$\text{So, } K = \{(15/1) \times 20\} = 300$$

Now,

$$L = \{(10^2 N) - 100\}$$

$$L = \{(100 \times 7) - 100\}$$

$$L = (700 - 100)$$

$$L = 600$$

$$\text{Therefore, the total number of children (boys + girls) living in each town} = (L/K * 100) = (600/300 * 100) = 200$$

Now, the total number of children (boys + girls) living in each town,

$$A = (2M - 5) \% \text{ of } 200 = (30 - 5) \% \text{ of } 200 = 25 \% \text{ of } 200 = 50$$

$$B = (M + N + 13) \% \text{ of } 200 = (15 + 7 + 13) \% \text{ of } 200 = 35 \% \text{ of } 200 = 70$$

$$C = M \% \text{ of } 200 = 15 \% \text{ of } 200 = 30$$

$$D = (2N + 4) \% \text{ of } 200 = (14 + 4) \% \text{ of } 200 = 18 \% \text{ of } 200 = 36$$

$$E = N \% = 7 \% \text{ of } 200 = 14$$

So, the total age of all the children (boys + girls) in each town,

$$A = (L + 50) = (600 + 50) = 650 \text{ years}$$

$$B = (K + L + 150) = (300 + 600 + 150) = 1050 \text{ years}$$

$$C = (K + 78) = (300 + 78) = 378 \text{ years}$$

$$D = (L - 264) = (600 - 264) = 336 \text{ years}$$

$$E = (K - 104) = (300 - 104) = 196 \text{ years}$$

Answer: A

According to the question,

Town A

The total number of children (boys + girls) living = 50

The total age of all the children (boys + girls) = 650

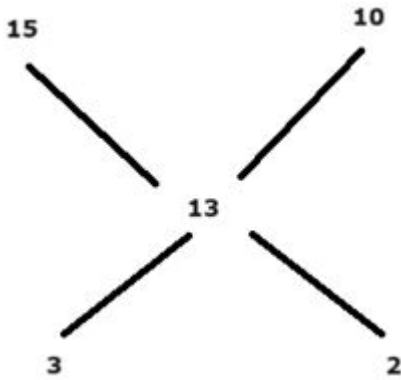
So, the average age of all the children (boys + girls) = $(650/50) = 13$

The average age of all the boys = 15 years

So, the average age of all the girls = $\{(15/150) \times 100\} = 10 \text{ years}$

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Using allegation method,



So, the ratio of the number of boys to that of girls = 3:2

Hence, the difference between the number of boys and girls living in town A = $\frac{1}{5}$ of 50 = 10.

2. Questions

$$2M - 5 + M + N + 13 + M + 2N + 4 + N = 100$$

$$M + N = 22 \text{ ----- (i)}$$

From the given equation, we have,

$$Z^2 - 13Z + 36 = 0$$

$$Z^2 - 9Z - 4Z + 36 = 0$$

$$Z(Z-9) - 4(Z-9) = 0$$

$$(Z-4)(Z-9) = 0$$

$$Z = 4 \text{ or } 9$$

So, the roots are = 4 and 9

Therefore, the value of N will be between 4 and 9.

So, the possible values of N = 5 or 7 (only prime numbers between 4 and 9)

Let, the value of N = 5

Now, put the value of N in equation (i), we get,

$$M + N = 22$$

$$M + 5 = 22$$

$$M = 17 \text{ (not possible)}$$

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Now, let the value of N = 7

Put the value of N in equation (i), we get,

$$M + N = 22$$

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$$M + 7 = 22$$

$$M = 15 \text{ (possible)}$$

15 is an odd composite number.

Therefore, the values of,

$$N = 7 \text{ (confirmed)}$$

$$M = 15 \text{ (confirmed)}$$

$$\text{So, } K = \{(15/1) \times 20\} = 300$$

Now,

$$L = \{(10^2 N) - 100\}$$

$$L = \{(100 \times 7) - 100\}$$

$$L = (700 - 100)$$

$$L = 600$$

$$\text{Therefore, the total number of children (boys + girls) living in each town} = (L/K * 100) = (600/300 * 100) = 200$$

Now, the total number of children (boys + girls) living in each town,

$$A = (2M - 5) \% \text{ of } 200 = (30 - 5) \% \text{ of } 200 = 25 \% \text{ of } 200 = 50$$

$$B = (M + N + 13) \% \text{ of } 200 = (15 + 7 + 13) \% \text{ of } 200 = 35 \% \text{ of } 200 = 70$$

$$C = M \% \text{ of } 200 = 15 \% \text{ of } 200 = 30$$

$$D = (2N + 4) \% \text{ of } 200 = (14 + 4) \% \text{ of } 200 = 18 \% \text{ of } 200 = 36$$

$$E = N \% = 7 \% \text{ of } 200 = 14$$

So, the total age of all the children (boys + girls) in each town,

$$A = (L + 50) = (600 + 50) = 650 \text{ years}$$

$$B = (K + L + 150) = (300 + 600 + 150) = 1050 \text{ years}$$

$$C = (K + 78) = (300 + 78) = 378 \text{ years}$$

$$D = (L - 264) = (600 - 264) = 336 \text{ years}$$

$$E = (K - 104) = (300 - 104) = 196 \text{ years}$$

Answer: D

Town B

The total number of children (boys + girls) living = 70

So, the number of boys living = $4/7$ of $70 = 40$

And, the number of girls living = $3/7$ of $70 = 30$

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Let, the average age of each boy and each girl be p years and $(p-14)$ years respectively.

So, the total age of all the boys = $(40 \times p) = 40p$ years

And, the total age of all the girls = $\{30 \times (p-14)\} = (30p-420)$ years

Now,

$$40p + (30p-420) = 1050$$

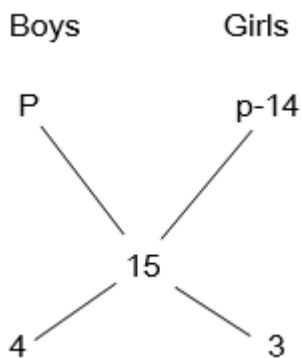
$$70p = 1470$$

$$p = 21$$

Hence, the sum of the ages of all the girls = $(30p-420) = (630-420) = 210$ years.

Alternate Method -

Using allegation method –



$$p - 15 / 15 - p + 14 = 4/3$$

$$p = 21$$

The total age of all the girls = $30 * (p - 14) = 30 * (21 - 14) = 30 * 7 = 210$ years

3. Questions

$$2M - 5 + M + N + 13 + M + 2N + 4 + N = 100$$

$$M + N = 22 \text{ ----- (i)}$$

From the given equation, we have,

$$Z^2 - 13Z + 36 = 0$$

$$Z^2 - 9Z - 4Z + 36 = 0$$

$$Z(Z-9) - 4(Z-9) = 0$$

$$(Z-4)(Z-9) = 0$$

$$Z = 4 \text{ or } 9$$

So, the roots are = 4 and 9

Therefore, the value of N will be between 4 and 9.

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So, the possible values of $N = 5$ or 7 (only prime numbers between 4 and 9)

Let, the value of $N = 5$

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15 is an odd composite number.

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$$N = 7 \text{ (confirmed)}$$

$$M = 15 \text{ (confirmed)}$$

$$\text{So, } K = \{(15/1) \times 20\} = 300$$

Now,

$$L = \{(10^2 N) - 100\}$$

$$L = \{(100 \times 7) - 100\}$$

$$L = (700 - 100)$$

$$L = 600$$

Therefore, the total number of children (boys + girls) living in each town = $(L/K \times 100) = (600/300 \times 100) = 200$

Now, the total number of children (boys + girls) living in each town,

$$A = (2M - 5) \% \text{ of } 200 = (30 - 5) \% \text{ of } 200 = 25 \% \text{ of } 200 = 50$$

$$B = (M + N + 13) \% \text{ of } 200 = (15 + 7 + 13) \% \text{ of } 200 = 35 \% \text{ of } 200 = 70$$

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So, the total age of all the children (boys + girls) in each town,

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$$D = (L-264) = (600-264) = 336 \text{ years}$$

$$E = (K-104) = (300-104) = 196 \text{ years}$$

Answer: B

Town C

The percentage of the total number of people (children + adults) living = 100%

And, the percentage of the total number of adults (males + females) living = 70%

So, the percentage of the total number of children (boys + girls) living = $(100-70)\% = 30\%$

$$30\% = 30$$

$$1\% = (30/30) = 1$$

$$100\% = (100 \times 1) = 100$$

Therefore, the total number of people (children + adults) living = 100

So, the total number of adults (males + females) living = 70% of the 100 = 70

Hence, the number of males living in town C = $5/7$ of 70 = 50.

4. Questions

$$2M - 5 + M + N + 13 + M + 2N + 4 + N = 100$$

$$M + N = 22 \text{ ----- (i)}$$

From the given equation, we have,

$$Z^2 - 13Z + 36 = 0$$

$$Z^2 - 9Z - 4Z + 36 = 0$$

$$Z(Z-9) - 4(Z-9) = 0$$

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$$M + N = 22$$

$$M + 7 = 22$$

$$M = 15 \text{ (possible)}$$

15 is an odd composite number.

Therefore, the values of,

$$N = 7 \text{ (confirmed)}$$

$$M = 15 \text{ (confirmed)}$$

$$\text{So, } K = \{(15/1) \times 20\} = 300$$

Now,

$$L = \{(10^2 N) - 100\}$$

$$L = \{(100 \times 7) - 100\}$$

$$L = (700 - 100)$$

$$L = 600$$

$$\text{Therefore, the total number of children (boys + girls) living in each town} = (L/K * 100) = (600/300 * 100) = 200$$

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$$D = (L - 264) = (600 - 264) = 336 \text{ years}$$

@QuizByMickey

$$E = (K-104) = (300-104) = 196 \text{ years}$$

Answer: E

Town F

Fractional value of 11.11% = $1/9$

So, the total number of children (boys + girls) living = $10/9$ of $36 = 40$

Let, the number of boys and girls living be s and t respectively.

Therefore, the total age of all the boys = $(s \times 8) = 8s$ years

And, the total age of all the girls = $\{t \times (8/2) \times 1\} = (t \times 4) = 4t$ years

Now,

$$s + t = 40 \dots\dots\dots (i)$$

$$8s - 4t = 140 \dots\dots\dots (ii)$$

Solving both the equations, we get,

$$8s + 8t = 320$$

$$8s - 4t = 140$$

$$12t = 180$$

$$t = 15$$

Now, put the value of t in equation (i), we get,

$$s + t = 40$$

$$s + 15 = 40$$

$$s = 25$$

Hence, the ratio of the number of boys to that of girls living in town F =

$$s:t = 25:15 = 5:3.$$

5. Questions

$$2M - 5 + M + N + 13 + M + 2N + 4 + N = 100$$

$$M + N = 22 \text{ ----- (i)}$$

From the given equation, we have,

$$Z^2 - 13Z + 36 = 0$$

$$Z^2 - 9Z - 4Z + 36 = 0$$

$$Z(Z-9) - 4(Z-9) = 0$$

$$(Z-4)(Z-9) = 0$$

$$Z = 4 \text{ or } 9$$

@QuizByMickey

So, the roots are = 4 and 9

Therefore, the value of N will be between 4 and 9.

So, the possible values of N = 5 or 7 (only prime numbers between 4 and 9)

Let, the value of N = 5

Now, put the value of N in equation (i), we get,

$$M + N = 22$$

$$M + 5 = 22$$

$$M = 17 \text{ (not possible)}$$

17 is a prime number, but not an odd composite number.

Now, let the value of N = 7

Put the value of N in equation (i), we get,

$$M + N = 22$$

$$M + 7 = 22$$

$$M = 15 \text{ (possible)}$$

15 is an odd composite number.

Therefore, the values of,

$$N = 7 \text{ (confirmed)}$$

$$M = 15 \text{ (confirmed)}$$

$$\text{So, } K = \{(15/1) \times 20\} = 300$$

Now,

$$L = \{(10^2 N) - 100\}$$

$$L = \{(100 \times 7) - 100\}$$

$$L = (700 - 100)$$

$$L = 600$$

Therefore, the total number of children (boys + girls) living in each town = $(L/K \times 100) = (600/300 \times 100) = 200$

Now, the total number of children (boys + girls) living in each town,

$$A = (2M - 5) \% \text{ of } 200 = (30 - 5) \% \text{ of } 200 = 25 \% \text{ of } 200 = 50$$

$$B = (M + N + 13) \% \text{ of } 200 = (15 + 7 + 13) \% \text{ of } 200 = 35 \% \text{ of } 200 = 70$$

$$C = M \% \text{ of } 200 = 15 \% \text{ of } 200 = 30$$

$$D = (2N + 4) \% \text{ of } 200 = (14 + 4) \% \text{ of } 200 = 18 \% \text{ of } 200 = 36$$

@QuizByMickey

$$E = N\% = 7\% \text{ of } 200 = 14$$

So, the total age of all the children (boys + girls) in each town,

$$A = (L+50) = (600+50) = 650 \text{ years}$$

$$B = (K+L+150) = (300+600+150) = 1050 \text{ years}$$

$$C = (K+78) = (300+78) = 378 \text{ years}$$

$$D = (L-264) = (600-264) = 336 \text{ years}$$

$$E = (K-104) = (300-104) = 196 \text{ years}$$

Answer: A

Town E

The total number of children (boys + girls) living = 14

Let, the number of boys and girls living be v and $(14-v)$ respectively.

Now,

$$v \times (14-v) = 49$$

$$14v - v^2 = 49$$

$$v^2 - 14v + 49 = 0$$

$$v^2 - 7v - 7v + 49 = 0$$

$$v(v-7) - 7(v-7) = 0$$

$$(v-7)(v-7) = 0$$

$$v = 7 \text{ or } 7$$

$$\text{So, } v = 7$$

Therefore, the number of boys living = $v = 7$

And, the number of girls living = $(14-v) = (14-7) = 7$

The sum of the ages of all the girls = $(14 \times 7) = 98$ years

And, the total ages of all the children (boys + girls) = 196

So, the sum of the ages of all the boys = $(196-98) = 98$ years

Hence, the average age of all the boys = $(98/7) = 14$ years.

6. Questions

Club P

The total number of players playing all the sports = $(A+20)$

So, the number of players playing football = $40\% \text{ of } (A+20) = (0.40A+8)$

@QuizByMickey

The difference between the number of players playing cricket and hockey (cricket-hockey) = 10% of (A+20) = (0.10A+2)

The number of players playing hockey = 80

So, the number of players playing cricket = {80 + (0.10A+2)} = (0.10A+82)

Now,

$$(0.40A+8) + 80 + (0.10A+82) = (A+20)$$

$$0.50A + 170 = A + 20$$

$$0.50A = 150$$

$$A = 300$$

So, the total number of players playing all the sports = (A+20) = (300+20) = 320

The number of players playing cricket = (0.10A+82) = (30+82) = 112

The number of players playing football = (0.40A+8) = (120+8) = 128

Club Q

The total number of players playing all the sports = 210

So, the number of players playing football = 2B% of 210 = 4.20B

The difference between the number of players playing cricket and hockey (cricket-hockey) = (4.20B-42)

The number of players playing hockey = [{(4.20B)/5} × 1] = 0.84B

So, the number of players playing cricket = {0.84B + (4.20B-42)} = (5.04B-42)

Now,

$$4.20B + 0.84B + (5.04B-42) = 210$$

$$10.08B - 42 = 210$$

$$10.08B = 252$$

$$B = 25$$

So, the number of players playing cricket = (5.04B-42) = 84

The number of players playing football = 4.20B = 105

The number of players playing hockey = 0.84B = 21

Club R

The total number of players playing all the sports = (C+90)

So, the number of players playing football = C% of (C+90) = {(C²/100)+0.90C} = {(10C²+900C)/1000}

Now,

$$\{(10C^2+900C)/1000\} = 90$$

@QuizByMickey

$$10C^2 + 900C = 90000$$

$$10C^2 + 900C - 90000 = 0$$

$$C^2 + 90C - 9000 = 0$$

$$C^2 + 150C - 60C - 9000 = 0$$

$$C(C+150) - 60(C+150) = 0$$

$$(C-60)(C+150) = 0$$

$$C = 60 \text{ or } -150$$

So, $C = 60$ (positive integer)

Therefore, the total number of players playing all the sports = $(C+90) = (60+90) = 150$

The number of players playing football = 90

So, the sum of the number of players playing cricket and hockey = $(150-90) = 60$

Let, the number of players playing cricket be $3n$

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = $\frac{2}{3}$ of $3n$
 $= 2n$

Therefore, the number of players playing hockey = $(3n-2n) = n$

Now,

$$(3n+n) = 60$$

$$4n = 60$$

$$n = 15$$

So, the number of players playing cricket = $3n = 45$

And, the number of players playing hockey = $n = 15$

Club S

The total number of players playing all the sports = 20D

So, the number of players playing football = 55% of 20D = 11D @QuizByMickey

The number of players playing cricket = 150% of 80 = 120

Let, the number of players playing hockey be m

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = m

Therefore, the number of players playing cricket = $(m+m) = 2m$

Now,

$$2m = 120$$

$$m = 60$$

So, the number of players playing hockey = $m = 60$

Now,

$$11D + 120 + 60 = 20D$$

$$9D = 180$$

$$D = 20$$

Hence, the total number of players playing all the sports = $20D = 400$

Club T

The total number of players playing all the sports = 240

So, the number of players playing football = $(6E+3) \% \text{ of } 240 = (14.40E+7.20)$

The number of players playing cricket = $3E$

Now,

$$(14.40E+7.20) = 180$$

$$14.40E = 172.80$$

$$E = 12$$

So, the number of players playing football = $(14.40E+7.20) = (172.80+7.20) = 180$

And, the number of players playing cricket = $3E = 36$

The difference between the number of players playing cricket and football (football-cricket) = $(180-36) = 144$

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = $\sqrt{144} = 12$.

Therefore, the number of players playing hockey = $(36-12) = 24$

Club	The total number of players playing	The number of players playing cricket	The number of players playing football	The number of players playing hockey
P	320	112	128	80
Q	210	84	105	21
R	150	45	90	15
S	400	120	220	60
T	240	36	180	24
Total	1320	397	723	200

Answer: D

According to the question,

Club P

The total number of players playing cricket = 112

Let, the average number of runs scored by 44 players and 38 players be $5r$ and $4r$ respectively.

So, the remaining number of players playing cricket = $\{112 - (44 + 38)\} = (112 - 82) = 30$

Therefore, the total number of runs scored by 30 players = $(30 \times 30) = 900$

And, the total number of runs scored by all the players playing cricket = 2760

So, the sum of the total number of runs scored by 44 players and 38 players = $(2760 - 900) = 1860$

Now, the total number of runs scored by 44 players = $(44 \times 5r) = 220r$

And, the total number of runs scored by 38 players = $(38 \times 4r) = 152r$

So,

$$220r + 152r = 1860$$

$$372r = 1860$$

$$r = 5$$

Therefore, the total number of runs scored by 44 players = $220r = 1100$

And, the total number of runs scored by 38 players = $152r = 760$

Hence, the required difference = $(1100 - 760) = 340$.

7. Questions

Club P

The total number of players playing all the sports = $(A + 20)$

So, the number of players playing football = 40% of $(A + 20) = (0.40A + 8)$

The difference between the number of players playing cricket and hockey (cricket-hockey) = 10% of $(A + 20) = (0.10A + 2)$

The number of players playing hockey = 80

So, the number of players playing cricket = $\{80 + (0.10A + 2)\} = (0.10A + 82)$

Now,

$$(0.40A + 8) + 80 + (0.10A + 82) = (A + 20)$$

$$0.50A + 170 = A + 20$$

$$0.50A = 150$$

$$A = 300$$

So, the total number of players playing all the sports = $(A + 20) = (300 + 20) = 320$

The number of players playing cricket = $(0.10A+82) = (30+82) = 112$

The number of players playing football = $(0.40A+8) = (120+8) = 128$

Club Q

The total number of players playing all the sports = 210

So, the number of players playing football = $2B\%$ of 210 = $4.20B$

The difference between the number of players playing cricket and hockey (cricket-hockey) = $(4.20B-42)$

The number of players playing hockey = $\left[\frac{(4.20B)}{5} \times 1\right] = 0.84B$

So, the number of players playing cricket = $\{0.84B + (4.20B-42)\} = (5.04B-42)$

Now,

$$4.20B + 0.84B + (5.04B-42) = 210$$

$$10.08B - 42 = 210$$

$$10.08B = 252$$

$$B = 25$$

So, the number of players playing cricket = $(5.04B-42) = 84$

The number of players playing football = $4.20B = 105$

The number of players playing hockey = $0.84B = 21$

Club R

The total number of players playing all the sports = $(C+90)$

So, the number of players playing football = $C\%$ of $(C+90) = \left\{\frac{C^2}{100}+0.90C\right\} = \left\{\frac{(10C^2+900C)}{1000}\right\}$

Now,

$$\left\{\frac{(10C^2+900C)}{1000}\right\} = 90$$

$$10C^2 + 900C = 90000$$

$$10C^2 + 900C - 90000 = 0$$

$$C^2 + 90C - 9000 = 0$$

$$C^2 + 150C - 60C - 9000 = 0$$

$$C(C+150) - 60(C+150) = 0$$

$$(C-60)(C+150) = 0$$

$$C = 60 \text{ or } -150$$

So, $C = 60$ (positive integer)

Therefore, the total number of players playing all the sports = $(C+90) = (60+90) = 150$

@QuizByMickey

The number of players playing football = 90

So, the sum of the number of players playing cricket and hockey = $(150-90) = 60$

Let, the number of players playing cricket be $3n$

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = $\frac{2}{3}$ of $3n$
 $= 2n$

Therefore, the number of players playing hockey = $(3n-2n) = n$

Now,

$$(3n+n) = 60$$

$$4n = 60$$

$$n = 15$$

So, the number of players playing cricket = $3n = 45$

And, the number of players playing hockey = $n = 15$

Club S

The total number of players playing all the sports = 20D

So, the number of players playing football = 55% of 20D = 11D

The number of players playing cricket = 150% of 80 = 120

Let, the number of players playing hockey be m

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = m

Therefore, the number of players playing cricket = $(m+m) = 2m$

Now,

$$2m = 120$$

$$m = 60$$

So, the number of players playing hockey = $m = 60$

Now,

$$11D + 120 + 60 = 20D$$

$$9D = 180$$

$$D = 20$$

Hence, the total number of players playing all the sports = $20D = 400$

Club T

The total number of players playing all the sports = 240

So, the number of players playing football = $(6E+3) \%$ of 240 = $(14.40E+7.20)$

@QuizByMickey

The number of players playing cricket = $3E$

Now,

$$(14.40E + 7.20) = 180$$

$$14.40E = 172.80$$

$$E = 12$$

$$\text{So, the number of players playing football} = (14.40E + 7.20) = (172.80 + 7.20) = 180$$

$$\text{And, the number of players playing cricket} = 3E = 36$$

$$\text{The difference between the number of players playing cricket and football (football-cricket)} = (180 - 36) = 144$$

$$\text{So, the difference between the number of players playing cricket and hockey (cricket-hockey)} = \sqrt{144} = 12.$$

$$\text{Therefore, the number of players playing hockey} = (36 - 12) = 24$$

Club	The total number of players playing	The number of players playing cricket	The number of players playing football	The number of players playing hockey
P	320	112	128	80
Q	210	84	105	21
R	150	45	90	15
S	400	120	220	60
T	240	36	180	24
Total	1320	397	723	200

Answer: B

Club U

$$\text{The total number of players playing all the sports} = (240 - 40) = 200$$

$$\text{So, the number of players playing cricket} = X\% \text{ of } 200 = 2X$$

Now, let the number of players playing football and hockey be $13Y$ and $7Y$ respectively.

$$2X + 13Y + 7Y = 200$$

$$2X + 20Y = 200$$

$$X + 10Y = 100 \dots\dots\dots (i)$$

And,

$$2X - 13Y = 35 \dots\dots\dots (ii)$$

Solving both the equations, we get,

$$2X + 20Y = 200$$

$$2X - 13Y = 35$$

$$33Y = 165$$

$$Y = 5$$

Now, put the value of Y in equation (i), we get,

$$X + 10Y = 100$$

$$X + 50 = 100$$

$$X = 50$$

Hence, the number of players playing hockey = $7Y = 35$.

8. Questions

Club P

The total number of players playing all the sports = $(A+20)$

So, the number of players playing football = 40% of $(A+20) = (0.40A+8)$

The difference between the number of players playing cricket and hockey (cricket-hockey) = 10% of $(A+20) = (0.10A+2)$

The number of players playing hockey = 80

So, the number of players playing cricket = $\{80 + (0.10A+2)\} = (0.10A+82)$

Now,

$$(0.40A+8) + 80 + (0.10A+82) = (A+20)$$

$$0.50A + 170 = A + 20$$

$$0.50A = 150$$

$$A = 300$$

So, the total number of players playing all the sports = $(A+20) = (300+20) = 320$

The number of players playing cricket = $(0.10A+82) = (30+82) = 112$

The number of players playing football = $(0.40A+8) = (120+8) = 128$

Club Q

The total number of players playing all the sports = 210

So, the number of players playing football = 2B% of 210 = $4.20B$

The difference between the number of players playing cricket and hockey (cricket-hockey) = $(4.20B-42)$

The number of players playing hockey = $\{[(4.20B)/5] \times 1\} = 0.84B$

So, the number of players playing cricket = $\{0.84B + (4.20B-42)\} = (5.04B-42)$

Now,

$$4.20B + 0.84B + (5.04B - 42) = 210$$

$$10.08B - 42 = 210$$

$$10.08B = 252$$

$$B = 25$$

$$\text{So, the number of players playing cricket} = (5.04B - 42) = 84$$

$$\text{The number of players playing football} = 4.20B = 105$$

$$\text{The number of players playing hockey} = 0.84B = 21$$

Club R

$$\text{The total number of players playing all the sports} = (C + 90)$$

$$\text{So, the number of players playing football} = C\% \text{ of } (C + 90) = \left\{ \left(\frac{C^2}{100} \right) + 0.90C \right\} = \left\{ \frac{(10C^2 + 900C)}{1000} \right\}$$

Now,

$$\left\{ \frac{(10C^2 + 900C)}{1000} \right\} = 90$$

$$10C^2 + 900C = 90000$$

$$10C^2 + 900C - 90000 = 0$$

$$C^2 + 90C - 9000 = 0$$

$$C^2 + 150C - 60C - 9000 = 0$$

$$C(C + 150) - 60(C + 150) = 0$$

$$(C - 60)(C + 150) = 0$$

$$C = 60 \text{ or } -150$$

$$\text{So, } C = 60 \text{ (positive integer)}$$

$$\text{Therefore, the total number of players playing all the sports} = (C + 90) = (60 + 90) = 150$$

$$\text{The number of players playing football} = 90$$

@QuizByMickey

$$\text{So, the sum of the number of players playing cricket and hockey} = (150 - 90) = 60$$

$$\text{Let, the number of players playing cricket be } 3n$$

$$\text{So, the difference between the number of players playing cricket and hockey (cricket-hockey)} = \frac{2}{3} \text{ of } 3n = 2n$$

$$\text{Therefore, the number of players playing hockey} = (3n - 2n) = n$$

Now,

$$(3n + n) = 60$$

$$4n = 60$$

$$n = 15$$

So, the number of players playing cricket = $3n = 45$

And, the number of players playing hockey = $n = 15$

Club S

The total number of players playing all the sports = $20D$

So, the number of players playing football = 55% of $20D = 11D$

The number of players playing cricket = 150% of $80 = 120$

Let, the number of players playing hockey be m

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = m

Therefore, the number of players playing cricket = $(m+m) = 2m$

Now,

$$2m = 120$$

$$m = 60$$

So, the number of players playing hockey = $m = 60$

Now,

$$11D + 120 + 60 = 20D$$

$$9D = 180$$

$$D = 20$$

Hence, the total number of players playing all the sports = $20D = 400$

Club T

The total number of players playing all the sports = 240

So, the number of players playing football = $(6E+3)\%$ of $240 = (14.40E+7.20)$

The number of players playing cricket = $3E$

Now,

$$(14.40E+7.20) = 180$$

$$14.40E = 172.80$$

$$E = 12$$

So, the number of players playing football = $(14.40E+7.20) = (172.80+7.20) = 180$

And, the number of players playing cricket = $3E = 36$

The difference between the number of players playing cricket and football (football-cricket) = $(180-36) =$

@QuizByMickey

144

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = $\sqrt{144} = 12$.

Therefore, the number of players playing hockey = $(36-12) = 24$

Club	The total number of players playing	The number of players playing cricket	The number of players playing football	The number of players playing hockey
P	320	112	128	80
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Total	1320	397	723	200

Answer: A

The total number of players playing football from club Q = 105

The total number of players playing football from club R = 90

Let, the total number of players playing football above 25 years of age from club Q and the number of players playing football below 25 of age years from club R be $12w$ and $7w$ respectively.

Also, let the total number of players playing football below 25 years of age from club Q and the number of players playing football above 25 years of age from club R be $9z$ and $11z$ respectively.

Now,

$$12w + 9z = 105 \dots\dots\dots (i)$$

$$7w + 11z = 90 \dots\dots\dots (ii)$$

Solving both the equations, we get,

$$84w + 63z = 735$$

$$84w + 132z = 1080$$

$$69z = 345$$

$$z = 5$$

Now, put the value of z in equation (i), we get,

$$12w + 9z = 105$$

$$12w + 45 = 105$$

$$12w = 60$$

@QuizByMickey

$$w = 5$$

Therefore, the total number of players playing football above 25 years of age = $(12w + 11z) = (60 + 55) = 115$

And, the total number of players playing football below 25 years of age = $(9z + 7w) = (45 + 35) = 80$

Hence, the required % = $\{(115/80) \times 100\} \% = 143.75\%$.

9. Questions

Club P

The total number of players playing all the sports = $(A + 20)$

So, the number of players playing football = 40% of $(A + 20) = (0.40A + 8)$

The difference between the number of players playing cricket and hockey (cricket-hockey) = 10% of $(A + 20) = (0.10A + 2)$

The number of players playing hockey = 80

So, the number of players playing cricket = $\{80 + (0.10A + 2)\} = (0.10A + 82)$

Now,

$$(0.40A + 8) + 80 + (0.10A + 82) = (A + 20)$$

$$0.50A + 170 = A + 20$$

$$0.50A = 150$$

$$A = 300$$

So, the total number of players playing all the sports = $(A + 20) = (300 + 20) = 320$

The number of players playing cricket = $(0.10A + 82) = (30 + 82) = 112$

The number of players playing football = $(0.40A + 8) = (120 + 8) = 128$

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So, the number of players playing football = 2B% of 210 = $4.20B$

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The number of players playing hockey = $[\{(4.20B)/5\} \times 1] = 0.84B$

So, the number of players playing cricket = $\{0.84B + (4.20B - 42)\} = (5.04B - 42)$

Now,

$$4.20B + 0.84B + (5.04B - 42) = 210$$

$$10.08B - 42 = 210$$

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$$B = 25$$

So, the number of players playing cricket = $(5.04B - 42) = 84$

The number of players playing football = $4.20B = 105$

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Club R

The total number of players playing all the sports = $(C+90)$

So, the number of players playing football = $C\%$ of $(C+90) = \{(C^2/100)+0.90C\} = \{(10C^2+900C)/1000\}$

Now,

$$\{(10C^2+900C)/1000\} = 90$$

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$$C^2 + 90C - 9000 = 0$$

$$C^2 + 150C - 60C - 9000 = 0$$

$$C(C+150) - 60(C+150) = 0$$

$$(C-60)(C+150) = 0$$

$$C = 60 \text{ or } -150$$

So, $C = 60$ (positive integer)

Therefore, the total number of players playing all the sports = $(C+90) = (60+90) = 150$

The number of players playing football = 90

So, the sum of the number of players playing cricket and hockey = $(150-90) = 60$

Let, the number of players playing cricket be $3n$

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = $2/3$ of $3n$
 $= 2n$

Therefore, the number of players playing hockey = $(3n-2n) = n$

Now,

$$(3n+n) = 60$$

$$4n = 60$$

$$n = 15$$

So, the number of players playing cricket = $3n = 45$

And, the number of players playing hockey = $n = 15$

Club S

@QuizByMickey

The total number of players playing all the sports = 20D

So, the number of players playing football = 55% of 20D = 11D

The number of players playing cricket = 150% of 80 = 120

Let, the number of players playing hockey be m

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = m

Therefore, the number of players playing cricket = (m+m) = 2m

Now,

$$2m = 120$$

$$m = 60$$

So, the number of players playing hockey = m = 60

Now,

$$11D + 120 + 60 = 20D$$

$$9D = 180$$

$$D = 20$$

Hence, the total number of players playing all the sports = 20D = 400

Club T

The total number of players playing all the sports = 240

So, the number of players playing football = (6E+3) % of 240 = (14.40E+7.20)

The number of players playing cricket = 3E

Now,

$$(14.40E+7.20) = 180$$

$$14.40E = 172.80$$

$$E = 12$$

So, the number of players playing football = (14.40E+7.20) = (172.80+7.20) = 180

And, the number of players playing cricket = 3E = 36

The difference between the number of players playing cricket and football (football-cricket) = (180-36) = 144

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = $\sqrt{144}$ = 12.

Therefore, the number of players playing hockey = (36-12) = 24

Club	The total number of players playing	The number of players playing cricket	The number of players playing football	The number of players playing hockey
P	320	112	128	80
Q	210	84	105	21
R	150	45	90	15
S	400	120	220	60
T	240	36	180	24
Total	1320	397	723	200

Answer: C

Club S

The total number of players playing cricket = 120

And, the total number of players playing hockey = 60

So,

$$120 - L = 60 + K$$

$$K + L = 60$$

Hence, the value of $\sqrt{(K+L+4)} = \sqrt{(60+4)} = \sqrt{64} = 8$.

10. Questions

Club P

The total number of players playing all the sports = $(A+20)$

So, the number of players playing football = 40% of $(A+20) = (0.40A+8)$

The difference between the number of players playing cricket and hockey (cricket-hockey) = 10% of $(A+20) = (0.10A+2)$

The number of players playing hockey = 80

So, the number of players playing cricket = $\{80 + (0.10A+2)\} = (0.10A+82)$

Now,

$$(0.40A+8) + 80 + (0.10A+82) = (A+20)$$

$$0.50A + 170 = A + 20$$

$$0.50A = 150$$

$$A = 300$$

So, the total number of players playing all the sports = $(A+20) = (300+20) = 320$

The number of players playing cricket = $(0.10A+82) = (30+82) = 112$

The number of players playing football = $(0.40A+8) = (120+8) = 128$

Club Q

The total number of players playing all the sports = 210

So, the number of players playing football = $2B\%$ of 210 = $4.20B$

The difference between the number of players playing cricket and hockey (cricket-hockey) = $(4.20B-42)$

The number of players playing hockey = $\left[\frac{(4.20B)}{5} \times 1\right] = 0.84B$

So, the number of players playing cricket = $\{0.84B + (4.20B-42)\} = (5.04B-42)$

Now,

$$4.20B + 0.84B + (5.04B-42) = 210$$

$$10.08B - 42 = 210$$

$$10.08B = 252$$

$$B = 25$$

So, the number of players playing cricket = $(5.04B-42) = 84$

The number of players playing football = $4.20B = 105$

The number of players playing hockey = $0.84B = 21$

Club R

The total number of players playing all the sports = $(C+90)$

So, the number of players playing football = $C\%$ of $(C+90) = \left\{\frac{C^2}{100}+0.90C\right\} = \left\{\frac{(10C^2+900C)}{1000}\right\}$

Now,

$$\left\{\frac{(10C^2+900C)}{1000}\right\} = 90$$

$$10C^2 + 900C = 90000$$

$$10C^2 + 900C - 90000 = 0$$

$$C^2 + 90C - 9000 = 0$$

$$C^2 + 150C - 60C - 9000 = 0$$

$$C(C+150) - 60(C+150) = 0$$

$$(C-60)(C+150) = 0$$

$$C = 60 \text{ or } -150$$

So, $C = 60$ (positive integer)

Therefore, the total number of players playing all the sports = $(C+90) = (60+90) = 150$

The number of players playing football = 90

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So, the sum of the number of players playing cricket and hockey = $(150-90) = 60$

Let, the number of players playing cricket be $3n$

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = $\frac{2}{3}$ of $3n$
 $= 2n$

Therefore, the number of players playing hockey = $(3n-2n) = n$

Now,

$$(3n+n) = 60$$

$$4n = 60$$

$$n = 15$$

So, the number of players playing cricket = $3n = 45$

And, the number of players playing hockey = $n = 15$

Club S

The total number of players playing all the sports = $20D$

So, the number of players playing football = 55% of $20D = 11D$

The number of players playing cricket = 150% of $80 = 120$

Let, the number of players playing hockey be m

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = m

Therefore, the number of players playing cricket = $(m+m) = 2m$

Now,

$$2m = 120$$

$$m = 60$$

So, the number of players playing hockey = $m = 60$

Now,

$$11D + 120 + 60 = 20D$$

$$9D = 180$$

$$D = 20$$

Hence, the total number of players playing all the sports = $20D = 400$

Club T

The total number of players playing all the sports = 240

So, the number of players playing football = $(6E+3) \%$ of $240 = (14.40E+7.20)$

The number of players playing cricket = $3E$

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Now,

$$(14.40E+7.20) = 180$$

$$14.40E = 172.80$$

$$E = 12$$

$$\text{So, the number of players playing football} = (14.40E+7.20) = (172.80+7.20) = 180$$

$$\text{And, the number of players playing cricket} = 3E = 36$$

$$\text{The difference between the number of players playing cricket and football (football-cricket)} = (180-36) = 144$$

$$\text{So, the difference between the number of players playing cricket and hockey (cricket-hockey)} = \sqrt{144} = 12.$$

$$\text{Therefore, the number of players playing hockey} = (36-12) = 24$$

Club	The total number of players playing	The number of players playing cricket	The number of players playing football	The number of players playing hockey
P	320	112	128	80
Q	210	84	105	21
R	150	45	90	15
S	400	120	220	60
T	240	36	180	24
Total	1320	397	723	200

Answer: B

Club T

$$\text{The total number of players playing hockey} = 24$$

$$\text{The sum of the ages of all the players playing hockey} = (24 \times J) = 24J \text{ years}$$

$$\text{The average age of all the players including their coach} = (J+0.80) \text{ years}$$

$$\text{So, the sum of the ages of all the players including their coach} = \{(24+1) \times (J+0.80)\} = (25J+20)$$

$$\text{Therefore, the age of their coach} = \{(25J+20)-24J\}$$

So,

$$\{(25J+20)-24J\} = 2J$$

$$25J + 20 = 26J$$

$$J = 20$$

$$\text{Hence, the value of } J^2 = 20^2 = 400.$$

11. Questions

Bank A

Let, the number of male officers working to that of female clerks working be $3x$ and x respectively.

And, the ratio of the number of male clerks working to that of female officers working = $6:7$. Let, it be $6y$ and $7y$ respectively.

Now, the total number of male (officers + clerks) employees working = 150

And, the total number of female (officers + clerks) employees working = $\{(150/150) \times 100\} = 100$

So,

$$3x + 6y = 150 \dots\dots\dots (i)$$

$$x + 7y = 100 \dots\dots\dots (ii)$$

Solving both the equations, we get,

$$3x + 6y = 150$$

$$3x + 21y = 300$$

$$15y = 150$$

$$y = 10$$

Now, put the value of y in equation (i), we get,

$$3x + 6y = 150$$

$$3x + 60 = 150$$

$$3x = 90$$

$$x = 30$$

So, the total number of male officers working = $3x = 90$

The total number of male clerks working = $6y = 60$

And, the total number of female officers working = $7y = 70$

The total number of female clerks working = $x = 30$

Employees	The number of officers working	The number of clerks working	The total number of employees working
Males	90	60	150
Females	70	30	100
Total	160	90	250

Bank B

The total number of male (officers + clerks) employees working = 140

The total number of female (officers + clerks) employees working = $(140 - 80) = 60$

The ratio of the number of male officers to that of female officers working = $250:100 = 5:2$. Let, it be $5m$ and $2m$ respectively.

Also, let the number of male and female clerks working be $2n$ and n respectively.

Now,

$$5m + 2n = 140 \dots\dots\dots(i)$$

$$2m + n = 60 \dots\dots\dots(ii)$$

Solving both the equations, we get,

$$10m + 4n = 280$$

$$10m + 5n = 300$$

$$n = 20$$

Now, put the value of n in equation (i), we get,

$$5m + 2n = 140$$

$$5m + 40 = 140$$

$$5m = 100$$

$$m = 20$$

So, the total number of male officers working = $5m = 100$

The total number of male clerks working = $2n = 40$

And, the total number of female officers working = $2m = 40$

The total number of female clerks working = $n = 20$

Employees	The number of officers working	The number of clerks working	The total number of employees working
Males	100	40	140
Females	40	20	60
Total	140	60	200

Answer: B

According to the question,

Bank A (Male Employees)

Let, the number of senior officers and junior clerks working be $8p$ and $7p$ respectively.

Also, the ratio of the number of junior officers to that of senior clerks working = $2:1$. Let, it be $2q$ and q respectively.

The total number of officers working = 90

The total number of clerks working = 60

Now,

$$8p + 2q = 90 \dots\dots\dots (i)$$

$$7p + q = 60 \dots\dots\dots (ii)$$

Solving both the equations, we get,

$$56p + 14q = 630$$

$$56p + 8q = 480$$

$$6q = 150$$

$$q = 25$$

Now, put the value of q in equation (i), we get,

$$8p + 2q = 90$$

$$8p + 50 = 90$$

$$8p = 40$$

$$p = 5$$

Quantity I: The number of senior officers working = $8p = 40$

And, the number of senior clerks working = $q = 25$

So, the required difference = $(40 - 25) = 15$

Hence, Quantity I = 15

Quantity II: The number of junior officers working = $2q = 50$

And, the number of junior clerks working = $7p = 35$

So, the required average = $\{(50 + 35)/2\} = (85/2) = 42.50$

Hence, Quantity II = 42.50

Therefore, Quantity I < Quantity II.

12. Questions

Bank A

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Let, the number of male officers working to that of female clerks working be $3x$ and x respectively.

And, the ratio of the number of male clerks working to that of female officers working = 6:7. Let, it be $6y$ and $7y$ respectively.

Now, the total number of male (officers + clerks) employees working = 150

And, the total number of female (officers + clerks) employees working = $\{(150/150) \times 100\} = 100$

So,

$$3x + 6y = 150 \dots\dots\dots (i)$$

$$x + 7y = 100 \dots\dots\dots (ii)$$

Solving both the equations, we get,

$$3x + 6y = 150$$

$$3x + 21y = 300$$

$$15y = 150$$

$$y = 10$$

Now, put the value of y in equation (i), we get,

$$3x + 6y = 150$$

$$3x + 60 = 150$$

$$3x = 90$$

$$x = 30$$

So, the total number of male officers working = $3x = 90$

The total number of male clerks working = $6y = 60$

And, the total number of female officers working = $7y = 70$

The total number of female clerks working = $x = 30$

Employees	The number of officers working	The number of clerks working	The total number of employees working
Males	90	60	150
Females	70	30	100
Total	160	90	250

Bank B

The total number of male (officers + clerks) employees working = 140

The total number of female (officers + clerks) employees working = $(140 - 80) = 60$

The ratio of the number of male officers to that of female officers working = $250:100 = 5:2$. Let, it be 5m and 2m respectively.

Also, let the number of male and female clerks working be 2n and n respectively.

Now,

$$5m + 2n = 140 \dots\dots\dots (i)$$

$$2m + n = 60 \dots\dots\dots (ii)$$

Solving both the equations, we get,

$$10m + 4n = 280$$

$$10m + 5n = 300$$

$$n = 20$$

Now, put the value of n in equation (i), we get,

$$5m + 2n = 140$$

$$5m + 40 = 140$$

$$5m = 100$$

$$m = 20$$

So, the total number of male officers working = $5m = 100$

The total number of male clerks working = $2n = 40$

And, the total number of female officers working = $2m = 40$

The total number of female clerks working = $n = 20$

Employees	The number of officers working	The number of clerks working	The total number of employees working
Males	100	40	140
Females	40	20	60
Total	140	60	200

Answer: A

Bank B (Female Clerks)

The total number of employees working = 20

The number of post graduates working = $4K$

The number of graduates working – The number of post graduates working = K^2

The number of graduates working – $4K = K^2$

So, the number of graduates working = $(4K + K^2)$

Now,

$$(4K + K^2) + 4K = 20$$

$$K^2 + 8K - 20 = 0$$

$$K^2 + 10K - 2K - 20 = 0$$

$$K(K+10) - 2(K+10) = 0$$

$$(K-2)(K+10) = 0$$

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$$K = 2 \text{ or } -10$$

$$K = 2 \text{ (positive integer)}$$

So,

$$2K = L$$

$$L = 4$$

Hence, the value of $\sqrt{L} = \sqrt{4} = 2$.

13. Questions

Bank A

Let, the number of male officers working to that of female clerks working be $3x$ and x respectively.

And, the ratio of the number of male clerks working to that of female officers working = $6:7$. Let, it be $6y$ and $7y$ respectively.

Now, the total number of male (officers + clerks) employees working = 150

And, the total number of female (officers + clerks) employees working = $\{(150/150) \times 100\} = 100$

So,

$$3x + 6y = 150 \dots\dots\dots (i)$$

$$x + 7y = 100 \dots\dots\dots (ii)$$

Solving both the equations, we get,

$$3x + 6y = 150$$

$$3x + 21y = 300$$

$$15y = 150$$

$$y = 10$$

Now, put the value of y in equation (i), we get,

$$3x + 6y = 150$$

$$3x + 60 = 150$$

$$3x = 90$$

$$x = 30$$

So, the total number of male officers working = $3x = 90$

The total number of male clerks working = $6y = 60$

And, the total number of female officers working = $7y = 70$

The total number of female clerks working = $x = 30$

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Employees	The number of officers working	The number of clerks working	The total number of employees working
Males	90	60	150
Females	70	30	100
Total	160	90	250

Bank B

The total number of male (officers + clerks) employees working = 140

The total number of female (officers + clerks) employees working = $(140 - 80) = 60$

The ratio of the number of male officers to that of female officers working = $250:100 = 5:2$. Let, it be $5m$ and $2m$ respectively.

Also, let the number of male and female clerks working be $2n$ and n respectively.

Now,

$$5m + 2n = 140 \dots\dots\dots(i)$$

$$2m + n = 60 \dots\dots\dots(ii)$$

Solving both the equations, we get,

$$10m + 4n = 280$$

$$10m + 5n = 300$$

$$n = 20$$

Now, put the value of n in equation (i), we get,

$$5m + 2n = 140$$

$$5m + 40 = 140$$

$$5m = 100$$

$$m = 20$$

So, the total number of male officers working = $5m = 100$

The total number of male clerks working = $2n = 40$

And, the total number of female officers working = $2m = 40$

The total number of female clerks working = $n = 20$

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Employees	The number of officers working	The number of clerks working	The total number of employees working
Males	100	40	140
Females	40	20	60
Total	140	60	200

Answer: C

Bank C

The total number of employees working = $\{(250+200)/2\} = (450/2) = 225$

The total number of clerks working = $4/3$ of $90 = 120$ (Fractional value of $33.33\% = 1/3$)

Hence, the total number of officers working in bank C = $(225-120) = 105$.

14. Questions

Answer: B

Equation I

$$X^2 - 41X + P^2 = 0$$

The roots are = 25 and 16

The sum of the roots = $(25+16) = 41$

$$5^2 = 25$$

$$4^2 = 16$$

Both the roots are squares of the positive integer.

So, a and b are the roots of the equation I

Therefore, a = 25

And, b = 16

So, the product of the roots = P^2

$$P^2 = (a \times b)$$

$$P^2 = (25 \times 16)$$

$$P^2 = 400$$

$$P = 20$$

Equation II

$$Y^2 - 35Y + 2Q = 0$$

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Roots of the equation = 25 and 10 ($c=25$, $d=10$, $\Rightarrow 25-10 = 15$) (it's not satisfied)

Roots of the equation = 30 and 5 ($c=30$, $d=5 \Rightarrow 30-5 = 25$) (it's also not satisfied)

So, Roots of the equation = 20 and 15 ($c=20$, $d=15 \Rightarrow 20-15 = 5$) (it's satisfied)

The sum of the roots = $(20+15) = 35$

Both 20 and 15 are exact multiples of 5.

So, c and d are the roots of the equation II

Therefore, $c = 20$

And, $d = 15$

So, the product of the roots = $2Q$

$$2Q = (c \times d)$$

$$2Q = (20 \times 15) = 300$$

$$Q = 150$$

Equation III

$$Z^2 - 34Z + (12R+1) = 0$$

The roots of the equation are equal

The sum of the roots = 34

So, both the roots = $(34/2) = 17$

17 is the prime number

Hence, e and f are the roots of the equation.

$$e = 17$$

$$f = 17$$

The product of the roots = $(12R+1)$

So,

$$(12R+1) = (e \times f)$$

$$12R = \{(17 \times 17) - 1\}$$

$$12R = (289 - 1)$$

$$12R = 288$$

$$R = 24$$

Now, the value of,

$$[(Q-50)/(R-P)] = [(150-50) / (24-20)] = (100/4) = 25.$$

15. Questions

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Answer: A

Equation I

$$A^2 + A - M = 0$$

The sum of the roots = -1

The roots of Y = 20 and -21

$$\{20 + (-21)\} = -1$$

So, Y is the roots of equation I

$$\text{The product of the roots} = \{20 \times (-21)\} = -420$$

$$\text{So, } M = 420$$

Equation II

$$B^2 - 2B - 3N = 0$$

The sum of the roots = 2

The roots of Z = -7 and 9

$$(-7+9) = 2$$

So, Z is the roots of the equation II

$$\text{The product of the roots} = (-7 \times 9) = -63$$

So,

$$3N = 63$$

$$N = 21$$

Equation III

$$C^2 - 53C + 2K = 0$$

The sum of the roots = 53

The roots of X = 30 and 23

$$(30+23) = 53$$

So, Z is the roots of the equation III.

$$\text{The product of the roots} = (30 \times 23) = 690$$

So,

$$2K = 690$$

$$K = 345$$

16. Questions

@QuizByMickey

Class I

The percentage of the number of boys who failed the examination = $X\%$

So, the percentage of the number of boys who passed the examination = $(100-X)\%$

$$(100-X)/X = 3/2$$

$$200 - 2X = 3X$$

$$5X = 200$$

$$X = 40$$

The ratio of the number of girls who passed to those of failed the examination = 3:1. (Fractional value of $66.67\% = 2/3$)

The percentage of the number of girls who passed the examination = $(Y+15)\%$

So, the percentage of the number of girls who failed the examination = $\{100-(Y+15)\} \% = (85-Y) \%$

$$(Y+15) / (85-Y) = 3/1$$

$$Y + 15 = 255 - 3Y$$

$$4Y = 240$$

$$Y = 60$$

Class I

The percentage of the number of boys who failed the examination = $X\% = 40\%$

The percentage of the number of boys who passed the examination = $(100-40) \% = 60\%$

The percentage of the number of girls who passed the examination = $(Y+15)\% = (60+15) \% = 75\%$

So, the percentage of the number of girls who failed the examination = $(85-Y) \% = (85-60) \% = 25\%$

Class II

The percentage of the number of boys who failed the examination = $(Y-15) \% = (60-15)\% = 45\%$

So, the percentage of the number of boys who passed the examination = $(100-45) \% = 55\%$

The percentage of the number of girls who passed the examination = $2X\% = 80\%$

So, the percentage of the number of girls who failed the examination = $(100-80) \% = 20\%$

Class III

The percentage of the number of boys who failed the examination = $(X-10) \% = (40-10) \% = 30\%$

So, the percentage of the number of boys who passed the examination = $(100-30) \% = 70\%$

The percentage of the number of girls who passed the examination = $(Y-5) \% = (60-5) \% = 55\%$

So, the percentage of the number of girls who failed the examination = $(100-55) \% = 45\%$

Class IV

The percentage of the number of boys who failed the examination = $(0.50Y-5)\% = (30-5)\% = 25\%$

So, the percentage of the number of boys who passed the examination = $(100-25)\% = 75\%$

The percentage of the number of girls who passed the examination = $1.50X\% = 60\%$

So, the percentage of the number of girls who failed the examination = $(100-60)\% = 40\%$

Answer: B

According to the question,

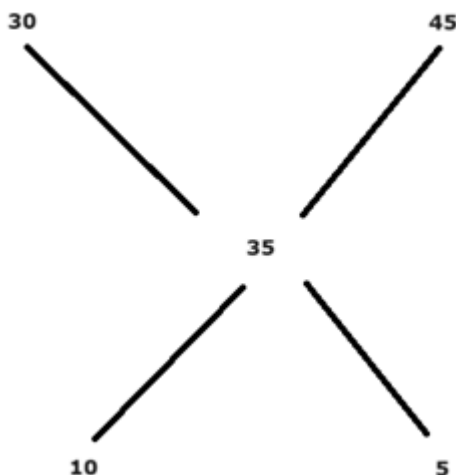
Class III

The percentage of the total number of students (boys + girls) who failed the examination = $(100-65)\% = 35\%$

The percentage of the number of boys who failed the examination = 30%

The percentage of the number of girls who failed the examination = 45%

Using allegation method,



So, the ratio of the number of boys to that of girls who appeared in the examination = $10:5 = 2:1$

Therefore, the total number of girls who appeared in the examination = $1/3$ of $360 = 120$

Hence, the number of girls who failed the examination = 45% of $120 = 54$.

17. Questions

Class I

The percentage of the number of boys who failed the examination = $X\%$

So, the percentage of the number of boys who passed the examination = $(100-X)\%$

$$(100-X)/X = 3/2$$

$$200 - 2X = 3X$$

$$5X = 200$$

$$X = 40$$

@QuizByMickey

The ratio of the number of girls who passed to those of failed the examination = 3:1. (Fractional value of 66.67% = 2/3)

The percentage of the number of girls who passed the examination = $(Y+15)\%$

So, the percentage of the number of girls who failed the examination = $\{100-(Y+15)\} \% = (85-Y) \%$

$$(Y+15) / (85-Y) = 3/1$$

$$Y + 15 = 255 - 3Y$$

$$4Y = 240$$

$$Y = 60$$

Class I

The percentage of the number of boys who failed the examination = $X\% = 40\%$

The percentage of the number of boys who passed the examination = $(100-40) \% = 60\%$

The percentage of the number of girls who passed the examination = $(Y+15)\% = (60+15) \% = 75\%$

So, the percentage of the number of girls who failed the examination = $(85-Y) \% = (85-60) \% = 25\%$

Class II

The percentage of the number of boys who failed the examination = $(Y-15) \% = (60-15)\% = 45\%$

So, the percentage of the number of boys who passed the examination = $(100-45) \% = 55\%$

The percentage of the number of girls who passed the examination = $2X\% = 80\%$

So, the percentage of the number of girls who failed the examination = $(100-80) \% = 20\%$

Class III

The percentage of the number of boys who failed the examination = $(X-10) \% = (40-10) \% = 30\%$

So, the percentage of the number of boys who passed the examination = $(100-30) \% = 70\%$

The percentage of the number of girls who passed the examination = $(Y-5) \% = (60-5) \% = 55\%$

So, the percentage of the number of girls who failed the examination = $(100-55) \% = 45\%$

Class IV

The percentage of the number of boys who failed the examination = $(0.50Y-5)\% = (30-5)\% = 25\%$

So, the percentage of the number of boys who passed the examination = $(100-25) \% = 75\%$

The percentage of the number of girls who passed the examination = $1.50X\% = 60\%$

So, the percentage of the number of girls who failed the examination = $(100-60) \% = 40\%$

Answer: A

Class II

The percentage of the number of boys who passed the examination = 55%

$$55\% = 88$$

$$1\% = (88/55) = 8/5$$

$$100\% = \{100 \times (8/5)\} = 160$$

So, the total number of boys who appeared in the examination = 160

The ratio of the total number of boys to that of girls who appeared in the examination = 8:15. Let, it be 8a and 15a respectively.

Now,

$$8a = 160$$

$$a = 20$$

Hence, the total number of students (boys + girls) who appeared in the examination from class II = $(8a+15a) = 23a = 460$.

18. Questions

Class I

The percentage of the number of boys who failed the examination = X%

So, the percentage of the number of boys who passed the examination = $(100-X)\%$

$$(100-X)/X = 3/2$$

$$200 - 2X = 3X$$

$$5X = 200$$

$$X = 40$$

The ratio of the number of girls who passed to those of failed the examination = 3:1. (Fractional value of $66.67\% = 2/3$)

The percentage of the number of girls who passed the examination = $(Y+15)\%$

So, the percentage of the number of girls who failed the examination = $\{100-(Y+15)\} \% = (85-Y) \%$

$$(Y+15) / (85-Y) = 3/1$$

$$Y + 15 = 255 - 3Y$$

$$4Y = 240$$

$$Y = 60$$

Class I

The percentage of the number of boys who failed the examination = $X\% = 40\%$

The percentage of the number of boys who passed the examination = $(100-40) \% = 60\%$

The percentage of the number of girls who passed the examination = $(Y+15)\% = (60+15) \% = 75\%$

So, the percentage of the number of girls who failed the examination = $(85-Y) \% = (85-60) \% = 25\%$

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Class II

The percentage of the number of boys who failed the examination = $(Y-15)\% = (60-15)\% = 45\%$

So, the percentage of the number of boys who passed the examination = $(100-45)\% = 55\%$

The percentage of the number of girls who passed the examination = $2X\% = 80\%$

So, the percentage of the number of girls who failed the examination = $(100-80)\% = 20\%$

Class III

The percentage of the number of boys who failed the examination = $(X-10)\% = (40-10)\% = 30\%$

So, the percentage of the number of boys who passed the examination = $(100-30)\% = 70\%$

The percentage of the number of girls who passed the examination = $(Y-5)\% = (60-5)\% = 55\%$

So, the percentage of the number of girls who failed the examination = $(100-55)\% = 45\%$

Class IV

The percentage of the number of boys who failed the examination = $(0.50Y-5)\% = (30-5)\% = 25\%$

So, the percentage of the number of boys who passed the examination = $(100-25)\% = 75\%$

The percentage of the number of girls who passed the examination = $1.50X\% = 60\%$

So, the percentage of the number of girls who failed the examination = $(100-60)\% = 40\%$

Answer: C

Class IV

The percentage of the number of boys who failed the examination = 25%

And the percentage of the number of boys who passed the examination = 75%

So,

$$(75-25)\% = 90$$

$$50\% = 90$$

$$1\% = (90/50) = 9/5$$

$$100\% = \{100 \times (9/5)\} = 180$$

So, the number of boys who failed the examination = 25% of $180 = 45$

Therefore, the number of girls who failed the examination = $(45+55) = 100$

The percentage of the number of girls who failed the examination = 40%

So,

$$40\% = 100$$

$$1\% = (100/40) = (5/2)$$

$$100\% = \{100 \times (5/2)\} = 250$$

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Therefore, the total number of girls who passed the examination = 250

Hence, the number of girls who passed the examination = 60% of 250 = 150.

19. Questions

Class I

The percentage of the number of boys who failed the examination = $X\%$

So, the percentage of the number of boys who passed the examination = $(100-X)\%$

$$(100-X)/X = 3/2$$

$$200 - 2X = 3X$$

$$5X = 200$$

$$X = 40$$

The ratio of the number of girls who passed to those of failed the examination = 3:1. (Fractional value of $66.67\% = 2/3$)

The percentage of the number of girls who passed the examination = $(Y+15)\%$

So, the percentage of the number of girls who failed the examination = $\{100-(Y+15)\} \% = (85-Y) \%$

$$(Y+15) / (85-Y) = 3/1$$

$$Y + 15 = 255 - 3Y$$

$$4Y = 240$$

$$Y = 60$$

Class I

The percentage of the number of boys who failed the examination = $X\% = 40\%$

The percentage of the number of boys who passed the examination = $(100-40) \% = 60\%$

The percentage of the number of girls who passed the examination = $(Y+15)\% = (60+15) \% = 75\%$

So, the percentage of the number of girls who failed the examination = $(85-Y) \% = (85-60) \% = 25\%$

Class II

The percentage of the number of boys who failed the examination = $(Y-15) \% = (60-15)\% = 45\%$

So, the percentage of the number of boys who passed the examination = $(100-45) \% = 55\%$

The percentage of the number of girls who passed the examination = $2X\% = 80\%$

So, the percentage of the number of girls who failed the examination = $(100-80) \% = 20\%$

Class III

The percentage of the number of boys who failed the examination = $(X-10) \% = (40-10) \% = 30\%$

So, the percentage of the number of boys who passed the examination = $(100-30) \% = 70\%$

The percentage of the number of girls who passed the examination = $(Y-5)\% = (60-5)\% = 55\%$

So, the percentage of the number of girls who failed the examination = $(100-55)\% = 45\%$

Class IV

The percentage of the number of boys who failed the examination = $(0.50Y-5)\% = (30-5)\% = 25\%$

So, the percentage of the number of boys who passed the examination = $(100-25)\% = 75\%$

The percentage of the number of girls who passed the examination = $1.50X\% = 60\%$

So, the percentage of the number of girls who failed the examination = $(100-60)\% = 40\%$

Answer: A

Class V

The total number of students (boys + girls) who appeared in the examination = 400

So, the number of boys who appeared in the examination = $5/8$ of 400 = 250

And, the number of girls who appeared in the examination = $3/8$ of 400 = 150

Now, the ratio of the number of boys who passed to that of girls who failed the examination = 6:1. Let, it be 6m and m respectively.

Also, of the number of boys who failed to that of girls who passed the examination = 7:12. Let, it be 7n and 12n respectively. (Fractional value of $41.67\% = 5/12$)

Now,

$$6m + 7n = 250 \dots\dots\dots(i)$$

And,

$$m + 12n = 150 \dots\dots\dots(ii)$$

Solving both the equations, we get,

$$6m + 7n = 250$$

$$6m + 72n = 900$$

$$65n = 650$$

$$n = 10$$

Now, put the value of n in equation (i), we get,

$$6m + 7n = 250$$

$$6m + 70 = 250$$

$$6m = 180$$

$$m = 30$$

Hence, the total number of students (boys + girls) who passed the examination = $(6m+12n) = (180+120) = 300$.

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20. Questions

Class I

The percentage of the number of boys who failed the examination = $X\%$

So, the percentage of the number of boys who passed the examination = $(100-X)\%$

$$(100-X)/X = 3/2$$

$$200 - 2X = 3X$$

$$5X = 200$$

$$X = 40$$

The ratio of the number of girls who passed to those of failed the examination = 3:1. (Fractional value of $66.67\% = 2/3$)

The percentage of the number of girls who passed the examination = $(Y+15)\%$

So, the percentage of the number of girls who failed the examination = $\{100-(Y+15)\} \% = (85-Y) \%$

$$(Y+15) / (85-Y) = 3/1$$

$$Y + 15 = 255 - 3Y$$

$$4Y = 240$$

$$Y = 60$$

Class I

The percentage of the number of boys who failed the examination = $X\% = 40\%$

The percentage of the number of boys who passed the examination = $(100-40) \% = 60\%$

The percentage of the number of girls who passed the examination = $(Y+15)\% = (60+15) \% = 75\%$

So, the percentage of the number of girls who failed the examination = $(85-Y) \% = (85-60) \% = 25\%$

Class II

The percentage of the number of boys who failed the examination = $(Y-15) \% = (60-15)\% = 45\%$

So, the percentage of the number of boys who passed the examination = $(100-45) \% = 55\%$

The percentage of the number of girls who passed the examination = $2X\% = 80\%$

So, the percentage of the number of girls who failed the examination = $(100-80) \% = 20\%$

Class III

The percentage of the number of boys who failed the examination = $(X-10) \% = (40-10) \% = 30\%$

So, the percentage of the number of boys who passed the examination = $(100-30) \% = 70\%$

The percentage of the number of girls who passed the examination = $(Y-5) \% = (60-5) \% = 55\%$

So, the percentage of the number of girls who failed the examination = $(100-55) \% = 45\%$

Class IV

The percentage of the number of boys who failed the examination = $(0.50Y - 5)\% = (30 - 5)\% = 25\%$

So, the percentage of the number of boys who passed the examination = $(100 - 25)\% = 75\%$

The percentage of the number of girls who passed the examination = $1.50X\% = 60\%$

So, the percentage of the number of girls who failed the examination = $(100 - 60)\% = 40\%$

Answer: D

Class I

The percentage of the number of boys who passed the examination = 60%

$$60\% = 120$$

$$1\% = (120/60) = 2$$

$$100\% = \{100 \times 2\} = 200$$

Therefore, the total number of boys who appeared in the examination = 200

And, the percentage of the number of girls who failed the examination = 25%

$$25\% = (120 - 85) = 35$$

$$1\% = (35/25) = (7/5)$$

$$100\% = \{100 \times (7/5)\} = 140$$

Therefore, the total number of girls who appeared in the examination = 140

So, the total number of students (boys + girls) who appeared in the examination = $(200 + 140) = 340$

The percentage of the total number students who appeared in the examination = $(100 - 15)\% = 85\%$

$$85\% = 340$$

$$1\% = (340/85) = 4$$

$$100\% = (100 \times 4) = 400$$

Hence, the total number of students (appeared + did not appear) in class I = 400.

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